

AC Power

Recall Phasors

$$\begin{aligned}v(t) &= V_m \cos(\omega t + \phi) \\&= \operatorname{Re}(V_m e^{j(\omega t + \phi)})\end{aligned}$$

$$v(t) = \operatorname{Re}(V_m e^{j\phi} e^{j\omega t})$$

$$v(t) = \operatorname{Re}(\bar{V} e^{j\omega t})$$

$$\bar{V} = V_m e^{j\phi} = V_m \angle \phi$$

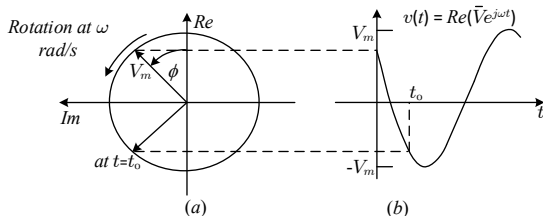


Figure: Representation of $\bar{V} e^{j\omega t}$: (a) phasor rotating counterclockwise, (b) its projection on the real axis, as a function of time.

$$v(t) = V_m \cos(\omega t + \phi)$$

$$\bar{V} = V_m \angle \phi$$

Recall Phasors (cont..)

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$\bar{V} = R\bar{I}$
L	$v = L \frac{di}{dt}$	$\bar{V} = j\omega L \bar{I}$
C	$i = C \frac{dv}{dt}$	$\bar{V} = \frac{\bar{I}}{j\omega C}$

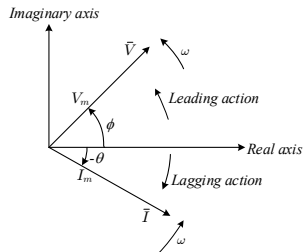


Figure: A phasor diagram showing $\bar{V} = V_m \angle \phi$ and $\bar{I} = I_m \angle -\theta$.

Recall RMS Value

The RMS value of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.

$$P = \frac{1}{T} \int_0^T i^2 R dt = \frac{R}{T} \int_0^T i^2 dt$$

$$P = I_{rms}^2 R$$

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2 dt}$$

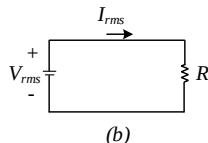
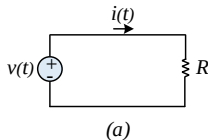


Figure: Finding the rms current: (a) ac circuit, (b) dc circuit.

$$X_{rms} = \sqrt{\frac{1}{T} \int_0^T x^2 dt}$$

Recall RMS Value (cont..)

$$i(t) = I_m \cos \omega t$$

$$\begin{aligned} I_{rms} &= \sqrt{\frac{1}{T} \int_0^T I_m^2 \cos^2 \omega t dt} \\ &= \sqrt{\frac{I_m^2}{T} \int_0^T \frac{1}{2} (1 + \cos 2\omega t) dt} = \frac{I_m}{\sqrt{2}} \end{aligned}$$

$$v(t) = V_m \cos \omega t$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

Instantaneous Power

$$p(t) = v(t)i(t)$$

The instantaneous power (in watts) is the power at any instant of time.

$$v(t) = V_m \cos(\omega t + \theta_v)$$

$$i(t) = I_m \cos(\omega t + \theta_i)$$

$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

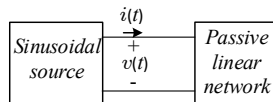


Figure: Sinusoidal source and passive linear circuit.

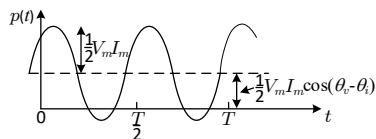


Figure: The instantaneous power $p(t)$ entering a circuit.

Average Power

The average power, in watts, is the average of the instantaneous power over one period.

$$P = \frac{1}{T} \int_0^T p(t) dt$$

$$\begin{aligned} P &= \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) dt + \\ &\frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) dt \\ &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \frac{1}{T} \int_0^T dt \\ &+ \frac{1}{2} V_m I_m \frac{1}{T} \int_0^T \cos(2\omega t + \theta_v + \theta_i) dt \end{aligned}$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Average Power (cont..)

$$\begin{aligned}\frac{1}{2}\bar{V}\tilde{I}^* &= \frac{1}{2}V_mI_m\angle\theta_v - \theta_i \\ &= \frac{1}{2}V_mI_m[\cos(\theta_v - \theta_i) + j\sin(\theta_v - \theta_i)]\end{aligned}$$

$$P = \frac{1}{2}\text{Re}[\bar{V}\tilde{I}^*] = \frac{1}{2}V_mI_m\cos(\theta_v - \theta_i)$$

A resistive load (R) absorbs power at all times, while a reactive load (L or C) absorbs zero average power.

Average Power (cont..)

The average power can be written in terms of the rms values.

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ &= V_{rms} I_{rms} \cos(\theta_v - \theta_i) \end{aligned}$$

The average power absorbed by a resistor R

$$P = I_{rms}^2 R = \frac{V_{rms}^2}{R}$$

Apparent Power

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i) = S \cos(\theta_v - \theta_i)$$

$$S = V_{rms} I_{rms}$$

The apparent power (in VA) is the product of the rms values of voltage and current. The apparent power is so called because it seems apparent that the power should be the voltage-current product, by analogy with dc resistive circuits. It is measured in volt-amperes or VA to distinguish it from the average or real power, which is measured in watts.

Power Factor

The power factor (pf) is the cosine of the phase difference between voltage and current.

$$pf = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

The power factor is dimensionless, since it is the ratio of the average power to the apparent power.

The angle $\theta_v - \theta_i$ is called the power factor angle.

pf is also the cosine of the angle of the load impedance.

$$\bar{Z} = \frac{\bar{V}}{\bar{I}} = \frac{V_m \angle \theta_v}{I_m \angle \theta_i} = \frac{V_m}{I_m} \angle \theta_v - \theta_i$$

Complex Power

$$\bar{S} = \frac{1}{2} \bar{V} \bar{I}^*$$

$$\bar{S} = \bar{V}_{rms} \bar{I}_{rms}^*$$

$$\bar{V}_{rms} = \frac{\bar{V}}{\sqrt{2}} = V_{rms} \angle \theta_v$$

$$\bar{I}_{rms} = \frac{\bar{I}}{\sqrt{2}} = I_{rms} \angle \theta_i$$

$$\begin{aligned} \bar{S} &= V_{rms} I_{rms} \angle \theta_v - \theta_i \\ &= V_{rms} I_{rms} \cos(\theta_v - \theta_i) + j V_{rms} I_{rms} \sin(\theta_v - \theta_i) \end{aligned}$$

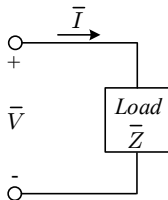


Figure: The voltage and current phasors associated with a load.

Complex Power (cont..)

$$\bar{Z} = \frac{\bar{V}}{\bar{I}} = \frac{\bar{V}_{rms}}{\bar{I}_{rms}} = \frac{V_{rms}}{I_{rms}} \angle \theta_v - \theta_i$$

$$\bar{V}_{rms} = \bar{Z} \bar{I}_{rms}$$

$$\bar{S} = I_{rms}^2 \bar{Z} = \frac{V_{rms}^2}{\bar{Z}^*} = \bar{V}_{rms} \bar{I}_{rms}^*$$

$$\bar{Z} = R + jX$$

$$\bar{S} = I_{rms}^2 (R + jX) = P + jQ$$

Complex Power (cont..)

$$P = \operatorname{Re}(\bar{S}) = I_{rms}^2 R$$

$$Q = \operatorname{Im}(\bar{S}) = I_{rms}^2 X$$

P is the average or real power and it depends on the loads resistance R . Q depends on the loads reactance X and is called the reactive power.

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i), \quad Q = V_{rms} I_{rms} \sin(\theta_v - \theta_i)$$

The real power P is the average power in watts delivered to a load; it is the only useful power. It is the actual power dissipated by the load. The reactive power Q is a measure of the energy exchange between the source and the reactive part of the load. The unit of Q is volt-ampere reactive (VAR) to distinguish it from the real power, whose unit is watt.

$Q = 0$ for resistive loads (unity pf).

$Q < 0$ for capacitive loads (leading pf).

$Q > 0$ for inductive loads (lagging pf).

Power Triangle

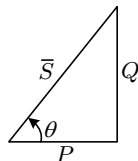


Figure: Power triangle.

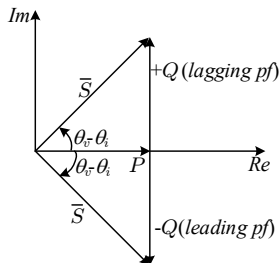
$$\begin{aligned}\text{Complex Power} = \bar{S} &= P + jQ = \bar{V}_{rms}(\bar{I}_{rms})^* \\ &= V_{rms}I_{rms} \angle \theta_v - \theta_i\end{aligned}$$

$$\text{Apparent Power} = S = |\bar{S}| = V_{rms}I_{rms} = \sqrt{P^2 + Q^2}$$

$$\text{Real Power} = P = \text{Re}(\bar{S}) = S \cos(\theta_v - \theta_i)$$

$$\text{Reactive Power} = Q = \text{Im}(\bar{S}) = S \sin(\theta_v - \theta_i)$$

$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$



Power Factor Correction

The process of increasing the power factor without altering the voltage or current to the original load is known as power factor correction.

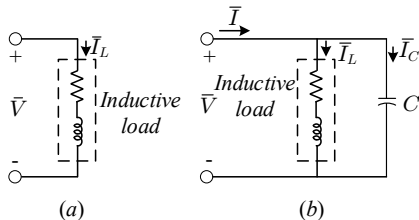


Figure: Power factor correction: (a) original inductive load, (b) inductive load with improved power factor.

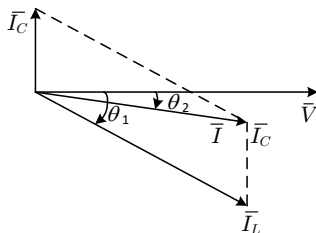


Figure: Phasor diagram showing the effect of adding a capacitor in parallel with the inductive load.

Power Factor Correction (cont..)

If the original inductive load has apparent power S_1 , then

$$P = S_1 \cos \theta_1, \quad Q = S_1 \sin \theta_1 = P \tan \theta_1$$

To increase the power factor without altering the real power (i.e., $P = S_2 \cos \theta_2$), the new reactive power is

$$Q_2 = P \tan \theta_2$$

The reduction in the reactive power is caused by the shunt capacitor.

$$Q_C = Q_1 - Q_2 = P(\tan \theta_1 - \tan \theta_2)$$

$$Q_C = \frac{V_{rms}^2}{X_C} = \omega C V_{rms}^2$$

$$C = \frac{Q_C}{\omega V_{rms}^2} = \frac{P(\tan \theta_1 - \tan \theta_2)}{\omega V_{rms}^2}$$

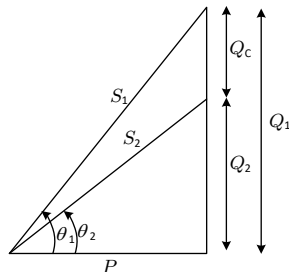


Figure: Power triangle illustrating power factor correction.

Power Measurement

The wattmeter is the instrument for measuring the average power.

$$P = |\bar{V}_{rms}| |\bar{I}_{rms}| \cos(\theta_v - \theta_i) = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$$

When the two coils are energized, the mechanical inertia of the moving system produces a deflection angle that is proportional to the average power.

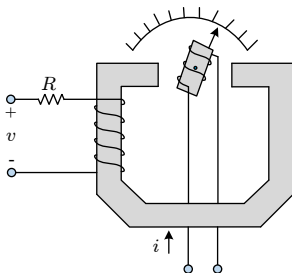


Figure: A wattmeter.

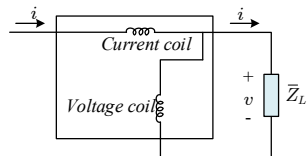


Figure: The wattmeter connected to the load.